

CelebA: IID / non-IID

COMPLETED VALIDATION COHORT - 10 shared seeds (91001-91010)

Accepted snapshot: 700/700 records, 2026-09-28T10:20:44 UTC

Data distribution			IID ($\alpha = 5000$)					non-IID ($\alpha = 5$)				
Category	Method	Metric	Benign	F-Flip	FedSA	S-DFA	Sp-DFA	Benign	F-Flip	FedSA	S-DFA	Sp-DFA
Vanilla FL	FedAvg	ACC (%) $\uparrow$	84.72 $\pm$ 11.74	89.57 $\pm$ 0.78	68.19 $\pm$ 5.77	66.01 $\pm$ 3.57	84.37 $\pm$ 2.38	86.85 $\pm$ 5.45	89.51 $\pm$ 1.82	74.76 $\pm$ 13.38	74.51 $\pm$ 13.37	85.48 $\pm$ 2.58
		AEOD $\downarrow$	0.0384 $\pm$ 0.0176	0.0519 $\pm$ 0.0096	0.1004 $\pm$ 0.0410	0.0950 $\pm$ 0.0477	0.0280 $\pm$ 0.0076	0.0428 $\pm$ 0.0137	0.0492 $\pm$ 0.0045	0.0644 $\pm$ 0.1097	0.0522 $\pm$ 0.0913	0.0267 $\pm$ 0.0157
		ASPD $\downarrow$	0.0937 $\pm$ 0.0366	0.1212 $\pm$ 0.0064	0.0993 $\pm$ 0.0380	0.0807 $\pm$ 0.0413	0.0693 $\pm$ 0.0161	0.0958 $\pm$ 0.0353	0.1161 $\pm$ 0.0143	0.0614 $\pm$ 0.0893	0.0591 $\pm$ 0.0756	0.0741 $\pm$ 0.0247
Fairness-aware	FairFed*	ACC (%) $\uparrow$	89.47 $\pm$ 0.85	89.79 $\pm$ 0.76	69.47 $\pm$ 4.65	62.59 $\pm$ 5.11	85.55 $\pm$ 2.31	89.59 $\pm$ 0.97	89.83 $\pm$ 0.92	71.45 $\pm$ 11.85	67.06 $\pm$ 11.91	86.18 $\pm$ 2.41
		AEOD $\downarrow$	0.0414 $\pm$ 0.0076	0.0489 $\pm$ 0.0042	0.1158 $\pm$ 0.0635	0.1594 $\pm$ 0.0822	0.0288 $\pm$ 0.0081	0.0388 $\pm$ 0.0057	0.0462 $\pm$ 0.0081	0.0759 $\pm$ 0.0860	0.1307 $\pm$ 0.1184	0.0328 $\pm$ 0.0133
		ASPD $\downarrow$	0.1102 $\pm$ 0.0077	0.1188 $\pm$ 0.0043	0.1120 $\pm$ 0.0554	0.1461 $\pm$ 0.0668	0.0781 $\pm$ 0.0116	0.1092 $\pm$ 0.0071	0.1163 $\pm$ 0.0071	0.0730 $\pm$ 0.0885	0.1171 $\pm$ 0.1067	0.0829 $\pm$ 0.0204
Robust FL	Median	ACC (%) $\uparrow$	86.28 $\pm$ 1.31	86.52 $\pm$ 1.35	84.66 $\pm$ 1.47	85.00 $\pm$ 1.38	85.54 $\pm$ 1.50	86.22 $\pm$ 1.31	86.51 $\pm$ 1.24	84.47 $\pm$ 1.30	84.48 $\pm$ 1.35	85.47 $\pm$ 1.26
		AEOD $\downarrow$	0.0510 $\pm$ 0.0081	0.0610 $\pm$ 0.0087	0.0438 $\pm$ 0.0135	0.0484 $\pm$ 0.0090	0.0534 $\pm$ 0.0096	0.0518 $\pm$ 0.0068	0.0594 $\pm$ 0.0084	0.0425 $\pm$ 0.0118	0.0461 $\pm$ 0.0142	0.0502 $\pm$ 0.0111
		ASPD $\downarrow$	0.0980 $\pm$ 0.0117	0.1062 $\pm$ 0.0111	0.0825 $\pm$ 0.0142	0.0852 $\pm$ 0.0127	0.0965 $\pm$ 0.0124	0.0963 $\pm$ 0.0096	0.1067 $\pm$ 0.0097	0.0799 $\pm$ 0.0113	0.0839 $\pm$ 0.0124	0.0929 $\pm$ 0.0086
Robust FL	FLTrust	ACC (%) $\uparrow$	89.43 $\pm$ 0.75	89.67 $\pm$ 0.82	89.68 $\pm$ 0.74	89.68 $\pm$ 0.74	89.72 $\pm$ 0.64	89.68 $\pm$ 0.60	89.83 $\pm$ 0.69	89.48 $\pm$ 0.70	89.48 $\pm$ 0.70	89.64 $\pm$ 0.71
		AEOD $\downarrow$	0.0432 $\pm$ 0.0075	0.0518 $\pm$ 0.0072	0.0422 $\pm$ 0.0069	0.0422 $\pm$ 0.0069	0.0496 $\pm$ 0.0061	0.0445 $\pm$ 0.0049	0.0493 $\pm$ 0.0054	0.0441 $\pm$ 0.0063	0.0441 $\pm$ 0.0063	0.0475 $\pm$ 0.0052
		ASPD $\downarrow$	0.1111 $\pm$ 0.0062	0.1207 $\pm$ 0.0064	0.1119 $\pm$ 0.0071	0.1119 $\pm$ 0.0071	0.1177 $\pm$ 0.0057	0.1132 $\pm$ 0.0060	0.1197 $\pm$ 0.0044	0.1120 $\pm$ 0.0041	0.1120 $\pm$ 0.0041	0.1153 $\pm$ 0.0053
Robust + fair	FairGuard*	ACC (%) $\uparrow$	89.18 $\pm$ 0.83	89.57 $\pm$ 1.24	49.32 $\pm$ 1.59	50.33 $\pm$ 1.72	83.20 $\pm$ 1.76	89.39 $\pm$ 1.11	89.84 $\pm$ 0.76	59.95 $\pm$ 16.51	61.29 $\pm$ 15.59	78.37 $\pm$ 14.94
		AEOD $\downarrow$	0.0317 $\pm$ 0.0026	0.0369 $\pm$ 0.0071	0.0042 $\pm$ 0.0132	0.0000 $\pm$ 0.0000	0.0205 $\pm$ 0.0098	0.0314 $\pm$ 0.0054	0.0394 $\pm$ 0.0103	0.0066 $\pm$ 0.0094	0.0108 $\pm$ 0.0161	0.0209 $\pm$ 0.0104
		ASPD $\downarrow$	0.0998 $\pm$ 0.0062	0.1078 $\pm$ 0.0092	0.0049 $\pm$ 0.0156	0.0000 $\pm$ 0.0000	0.0491 $\pm$ 0.0199	0.1026 $\pm$ 0.0086	0.1109 $\pm$ 0.0090	0.0169 $\pm$ 0.0251	0.0206 $\pm$ 0.0317	0.0552 $\pm$ 0.0238
Robust + fair	FLTrust+FairGuard*	ACC (%) $\uparrow$	89.93 $\pm$ 1.10	90.37 $\pm$ 0.60	77.83 $\pm$ 12.15	75.99 $\pm$ 16.52	86.44 $\pm$ 5.76	89.69 $\pm$ 1.68	90.07 $\pm$ 1.34	81.90 $\pm$ 11.66	80.44 $\pm$ 12.06	79.69 $\pm$ 13.94
		AEOD $\downarrow$	0.0365 $\pm$ 0.0075	0.0363 $\pm$ 0.0086	0.0298 $\pm$ 0.0299	0.0293 $\pm$ 0.0146	0.0368 $\pm$ 0.0144	0.0357 $\pm$ 0.0055	0.0426 $\pm$ 0.0062	0.0400 $\pm$ 0.0214	0.0359 $\pm$ 0.0262	0.0408 $\pm$ 0.0124
		ASPD $\downarrow$	0.1074 $\pm$ 0.0113	0.1118 $\pm$ 0.0100	0.0607 $\pm$ 0.0354	0.0652 $\pm$ 0.0419	0.0903 $\pm$ 0.0325	0.1085 $\pm$ 0.0102	0.1144 $\pm$ 0.0088	0.0851 $\pm$ 0.0350	0.0763 $\pm$ 0.0174	0.0892 $\pm$ 0.0276
Ours	GuardFed-AD2+	ACC (%) $\uparrow$	88.26 $\pm$ 1.04	88.39 $\pm$ 0.63	88.47 $\pm$ 0.91	88.69 $\pm$ 0.81	88.38 $\pm$ 0.89	88.59 $\pm$ 1.17	88.44 $\pm$ 0.74	88.41 $\pm$ 0.92	88.22 $\pm$ 0.78	88.34 $\pm$ 0.96
		AEOD $\downarrow$	0.0097 $\pm$ 0.0075	0.0107 $\pm$ 0.0096	0.0061 $\pm$ 0.0042	0.0077 $\pm$ 0.0046	0.0165 $\pm$ 0.0072	0.0070 $\pm$ 0.0042	0.0100 $\pm$ 0.0070	0.0078 $\pm$ 0.0056	0.0126 $\pm$ 0.0093	0.0087 $\pm$ 0.0066
		ASPD $\downarrow$	0.0625 $\pm$ 0.0138	0.0607 $\pm$ 0.0103	0.0640 $\pm$ 0.0142	0.0638 $\pm$ 0.0070	0.0559 $\pm$ 0.0156	0.0651 $\pm$ 0.0090	0.0589 $\pm$ 0.0092	0.0616 $\pm$ 0.0141	0.0584 $\pm$ 0.0189	0.0596 $\pm$ 0.0146

All 70 method/distribution/scenario cells use the same 10 seeds. Seed subsets follow the frozen protocol; all cells within each table use identical seeds.

Round 70; validation only (19,867 images). Mean $\pm$ sample SD (ddof = 1); ACC in %, gaps on [0, 1].

AEOD denotes the implemented absolute TPR gap. No significance claim follows from mean ranks alone.

* Project adaptations. GuardFed includes training-root group calibration; baseline outputs are uncalibrated.

Recipes were selected on non-IID Benign/S-DFA, including seed 91001; this is coverage transfer, not IID-specific tuning.

This table uses 686 cu128 and 14 cu130 records; the source cohort contains 14 cu130 records. No 70-round equivalence claim.

Stage A covers seven implementations; ten further manuscript baselines and mechanism controls remain outstanding.